

AWS VPC

1. Create VPC with 2 private and 2 public subnets.

Create a new vpc and in that added 2 public and 2 private subnets

Subnets (4) [Info](#) Last updated 2 minutes ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

Public-Subnet-2 [×](#) Public-Subnet-1 [×](#) Private-Subnet-1 [×](#) (+1) [Clear filters](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Public-Subnet-2	subnet-0a454875e5a922619	Available	vpc-01e8ea44337907063 MyV...	Off	10.0.2.0/24
☐	Public-Subnet-1	subnet-0ba5a9aa281715d09	Available	vpc-01e8ea44337907063 MyV...	Off	10.0.1.0/24
☐	Private-Subnet-1	subnet-08b1596d62f03f97e	Available	vpc-01e8ea44337907063 MyV...	Off	10.0.3.0/24
☐	Private-Subnet-2	subnet-083c59ed389e86b7f	Available	vpc-01e8ea44337907063 MyV...	Off	10.0.4.0/24

2. Enable DNS Hostname in VPC.

After creating VPC in action edit vpc settings change DNS Hostname

DNS settings

☒ Enable DNS resolution [Info](#)

☒ Enable DNS hostnames [Info](#)

VPC dashboard AWS Global View Filter by VPC: ▼ Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only Internet gateways	vpc-01e8ea44337907063 / MyVPC Details Info VPC ID vpc-01e8ea44337907063 DNS resolution Enabled Main network ACL acl-02b76df9c37ff94ba IPv6 CIDR (Network border group) - State Available Tenancy default Default VPC No Network Address Usage metrics Disabled Block Public Access Off DHCP option set dopt-0c4b0af3221994469 IPv4 CIDR 10.0.0.0/16 Route 53 Resolver DNS Firewall rule groups DNS hostnames Enabled Main route table rtb-001548248410c652b IPv6 pool - Owner ID 928430096503
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3. Enable Auto Assign Public IP in 2 public subnets.

1. Select public subnet 1 in actions open edit subnet settings

Subnets (1/4) [Info](#)

Find subnets by attribute or tag

Name = Public-Subnet-1 X Name = Public-Subnet-2 X Name = Private-Subnet-1 X (+1) [Clear filters](#)

	Name	Subnet ID	State	VPC
☐	Public-Subnet-2	subnet-0a454875e5a922619	Available	vpc-01e8ea44337
☒	Public-Subnet-1	subnet-0ba5a9aa281715d09	Available	vpc-01e8ea44337
☐	Private-Subnet-1	subnet-08b1596d62f03f97e	Available	vpc-01e8ea44337
☐	Private-Subnet-2	subnet-083e59edf389e86b7f	Available	vpc-01e8ea44337

Last updated 21 minutes ago [Actions](#)

- View details
- Create flow log
- Edit subnet settings
- Edit IPv6 CIDRs
- Edit network ACL association
- Edit route table association
- Edit CIDR reservations
- Share subnet

2. Enable auto sub assign public IPv4 address and save it

Edit subnet settings [Info](#)

Subnet

Subnet ID

[subnet-0ba5a9aa281715d09](#)

Name

[Public-Subnet-1](#)

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)

Option disabled because no customer owned pools found.

3. Same as public subnet 1 open select public subnet 2 and open actions in that edit subnet settings.

Find subnets by attribute or tag

Name = Public-Subnet-1 X Name = Public-Subnet-2 X Name = Private-Subnet-1 X (+1) [Clear filters](#)

	Name	Subnet ID	State	VPC
☒	Public-Subnet-2	subnet-0a454875e5a922619	Available	vpc-01e8ea44337
☐	Public-Subnet-1	subnet-0ba5a9aa281715d09	Available	vpc-01e8ea44337

view details

- Create flow log
- Edit subnet settings
- Edit IPv6 CIDRs
- Edit network ACL association
- Edit route table association

4. Enable auto sub assign public IPv4 address and save it

Edit subnet settings [Info](#)

Subnet

Subnet ID

[subnet-0a454875e5a922619](#)

Name

[Public-Subnet-2](#)

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

5. In Auto assign public IPv4 should show yes

Default subnet	Auto-assign public IPv4 add...	
No	Yes	↑
No	Yes	↑
No	No	↑
No	No	↑

4. Add 2 private subnets in private route table.

- Created new route table

Route tables (7) Info						
Find route tables by attribute or tag						
☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	project-rtb-private1-us-east-1a	rtb-08bb5187d24922e7c	subnet-0c4e3052806f6df...	-	No	vpc-01a1340e4a6a3309a proj.
☐	-	rtb-001548248410c652b	-	-	Yes	vpc-01e8ea44337907063 MyV

- Created new route table and given name as Private-RT

Route tables (1/1) Info	
Find route tables by attribute or tag	
Name = Private-RT	Clear filters
☒	Name
☒	Private-RT

- IN Subnet associations private-1 and private -2 subnets added

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Details

info

Route table ID

rtb-0ba084c9867b8e07b

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

VPC

vpc-01e8ea44337907063 | MyVPC

Owner ID

928430096503

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Private-Subnet-1	subnet-08b1596d62f03f97e	10.0.3.0/24	-
Private-Subnet-2	subnet-083c59ed389e86b7f	10.0.4.0/24	-

5. Add 2 public subnets in public route table.

- Created a new route table for public subnets

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - *optional*
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

- Given name as Public-RT and save it
- In routes added new internet gateway

Routes	Subnet associations	Edge associations	Route propagation	Tags
Routes (2) Both				
Filter routes				
Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0604070e37ed1a657	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route

- We need to create new internet gateway for that in vpc and given MyIGW

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

igw-0604070e37ed1a657 / MyIGW

Details [Info](#)

Internet gateway ID
[igw-0604070e37ed1a657](#)

State
Attached

VPC ID
[vpc-01e8ea44337907063 | MyVPC](#)

Tags (1)

Key	Value
Name	MyIGW

- Add in subnet associations both public subnets and save it

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
☒	Public-Subnet-2	subnet-0a454875e5a922619	10.0.2.0/24	–
☐	Private-Subnet-1	subnet-08b1596d62f03f97e	10.0.3.0/24	–
☒	Public-Subnet-1	subnet-0ba5a9aa281715d09	10.0.1.0/24	–
☐	Private-Subnet-2	subnet-083c59ed389e86b7f	10.0.4.0/24	–

6. Public route table will have the routes to internet and local.

Route tables (1/2) [Info](#)

Find route tables by attribute or tag

Name = Public-RT X Name = Private-RT X Clear filters

	Name	Route table ID	Explicit subnet associ...	Edge
☐	Private-RT	rtb-0ba084c9867b8e07b	2 subnets	-
☒	Public-RT	rtb-0986329a1857721cc	2 subnets	-

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

Routes (2) [Both](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0604070e37ed1a657	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route

7. Create EC2 in public subnet with t2.micro and install PHP.

Name and tags [Info](#)

Name

Public-EC2

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

[Recents](#) | [Quick Start](#)

Amazon Linux 	macOS 	Ubuntu 	Windows 	Red Hat 	SUSE Linux 	Debian 
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[Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

Free tier

Additional costs apply for AMIs with pre-installed software

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the

Key pair name - *required*

aws

Info

Info

vpc-01e8ea44337907063 (MyVPC)

10.0.0.0/16

Info

subnet-0ba5a9aa281715d09

Public-Subnet-1

VPC: vpc-01e8ea44337907063 Owner: 928430096503 Availability Zone: us-east-1a (use1-az1)

Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Info

Enable

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- Create security group

☐ Select existing security group

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ec2-user@ec2-98-92-188-179.compute-1.amazonaws.com
The authenticity of host 'ec2-98-92-188-179.compute-1.amazonaws.com (98.92.188.179)' can't be e
ED25519 key fingerprint is SHA256:DHGnhQHdTK/GrvrUKrPrg7Ti+S26dr55j3jSzFp7mc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-98-92-188-179.compute-1.amazonaws.com' (ED25519) to the list of
ts.
```

```
#_
~\####_      Amazon Linux 2023
~\#####\_
~\###|
~\#/___#'_-> https://aws.amazon.com/linux/amazon-linux-2023
~V~'
~
~..-.-/_
~\_/m/'
```

```
[ec2-user@ip-10-0-1-36 ~]$ sudo su -
[root@ip-10-0-1-36 ~]#
```

```
root@ip-10-0-1-36 ~]# sudo yum update -y
amazon Linux 2023 Kernel Livepatch repository
dependencies resolved.
Nothing to do.
Complete!
root@ip-10-0-1-36 ~]#
```

```
[root@ip-10-0-1-36 ~]# sudo yum install httpd php -y
Last metadata expiration check: 0:00:27 ago on Wed Apr 15 12:44:47 2026.
Dependencies resolved.
```

Package	Architecture	Version	Repository	S
Installing:				
httpd	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	4
php8.5	x86_64	8.5.4-1.amzn2023.0.1	amazonlinux	1

```
Complete!
[root@ip-10-0-1-36 ~]# sudo systemctl start httpd
[root@ip-10-0-1-36 ~]# sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/sys
.service.
[root@ip-10-0-1-36 ~]#
```

```
[root@ip-10-0-1-36 ~]# echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/index.php
<?php phpinfo(); ?>
[root@ip-10-0-1-36 ~]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Wed 2026-04-15 12:45:42 UTC; 2min 57s ago
     Docs: man:httpd.service(8)
   Main PID: 25828 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1014)
   Memory: 12.5M
```

PHP Version 8.5.4

System	Linux ip-10-0-1-36 ec2.internal 6.1.166-197.305.amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Mon Mar 23 09:53:26 UTC 2026 x86_64
Build Date	Mar 10 2026 23:15:23
Build System	Linux
Build Provider	Amazon Linux
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini

8. Configure NAT gateway in public subnet and connect to private instance.

NAT gateways [Info](#) Actions Create NAT gateway

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID
Name - optional Create a tag with a key of 'Name' and a value that you specify. MyNAT The name can be up to 256 characters long. Availability mode Info Choose whether to deploy across all zones in the region or restrict to a single availability zone. ☒ Regional - new Scales automatically across all regional AZs, simplifying management for multi AZ deployments. ☐ Zonal Provides granular control within a specific availability zone, adhering to subnet level settings. VPC Select a VPC in which to create the regional NAT gateway. vpc-01e8ea44337907063 (MyVPC) Connectivity type Select a connectivity type for the NAT gateway. ☒ Public						

rtb-0ba084c9867b8e07b / Private-RT

Details [Info](#)

Route table ID

rtb-0ba084c9867b8e07b

VPC

vpc-01e8ea44337907063 | MyVPC

Main

No

Owner ID

928430096503

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route
0.0.0.0/0	nat-131f6c1d0a771d508	Active	No	Crea
10.0.0.0/16	local	Active	No	Crea

Name and tags [Info](#)

Name

Private-EC2

▼ Application and OS Images (Amazon M

An AMI contains the operating system, application server, an field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application a

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair.

Key pair name - required

aws

▼

▼ Network settings Info

VPC - required Info

vpc-01e8ea44337907063 (MyVPC)

10.0.0.0/16

Subnet Info

subnet-08b1596d62f03f97e

Private-Subnet-1

VPC: vpc-01e8ea44337907063

Owner: 928430096503

Availability Zone: us-east-1a (use1-az1)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 10.0.3.0/24

```

runsaï@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ cat aws.pem
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAwszmZfP9T1Gjy+0A0wfgbb1N0VikrE3KrjaEg3y/j8IPvnnQr
tKnX79fAqJoh8JfegZ1h0dxIR/8zo/kxJQ+H5jgB3M+X5R2Rrsga7iqgdme5VGr
P6ngJYroJCwyr+TmGKEKwSdUhtuTHfrqucsZ/X0vt1nXg4zHioUA+XPYajCtJ7
0HocD2X9I/8CZ0eR01qbWz7Zq9Ng10WpNI73vXATZBLAX+oFXkg32VlJZ0QrAh1
VANb0e5jpyPqcPIMSJE5ISAMcy8i+IHGxgHQkU3AIWzJ1sbSyQ5pUVfBcZKZMQW
28oFB6rE+HUAz0Uqb2Iw2QzFfqD1N5Bjk5+wIDAQABAoIBACDDy5v7kjd65dTD
qPb6zT4W9601bMYXB3vnA5M1FdwY62eyGIKHTeooK1fHegaQeXjBR2oTl3myc9I
D7/DGRkdRiMt2+FK1fP5y6F4C88uvTSjNy30ygVLPNPFqV/IbZKmSZZ/PtREt9i
01aA3xtDguJZlUFnf/SNBWqe8hjjeJZML/I7pAT4xQC1YZtVe5h62T6wMXRuBVO
pecMnEm1lj8d8GxaIDq40wx+yFMZvTwvKrQsYBBtjNrn4wBJOpUEvdXMdyuMucn
UCe6CtNFw6qU+65NNQmCVVy+m6XTgRW4IWeV+kdaQfLyxiSx9TEdRhZ2vDlZrGe
n4SADECgYEA57e/14Ff8DCTvQ0BAZnVE1x5yX9KXy6/Nn6mIZrJCh2Y2pLLKr6F
PzMntVfPnNhtxvoPFeA3sw7qPYJniu0E4QDMBCtiCo+3kYn8IiQQIIUISv7Kf1n
QKqKQm1gVvgXqVHBipM1fjrW0Adi/12eGwakGsJBsuJBGjrtfCXKS8CgYEA1zM9
1S5SeaEATzs3W25ZzFtdXzmjVjUT7WI/sUGoRQsCYspr0AA4+bp47swIFDq03F7
sYQFtphvZkKhGpsa+zm1tG08bqkThrmY6BHNSuNhpNHekbdoHwCCqgKRxlLyAMA
921D32+0th83TZgRE1HnusuNRGw5N36bs4KMPUCgYEAwi/B088WIwoJud0oJz1N
b3r6iscnS8lM3ujqgHXfPy0vmfKj3W/A5/fszRjxmpnd1aR1w/V1eD+FmiqQr1p
X1eVozgn3p/2TKeafvQ9N9LFKYxRE0ysOuEwk+63MBNPUPxM397im3kHUUKPcu/
seRs3mfX0wJYVLAfMcbps0CgYB7Yt2HQqkTxh+Zthz50Jg9GpG1QJTQJkZVDXjm
oDain5Vwn0iqazY/uTLtd/6J2r8o0xKSFwhRMUVuFHxak2P89zG/pA++QyL29CM
rDWMg+LQZaR/V+Jp5ZQFdkOuupAw7gQ2Gwf5DEtIwM0sqCvIL+v1pg/xUzvbHbS
UBmoQKBgQDKRtXX+fN+RgZTL0Jf5Zi4Hq5zQXSgurgY4203i856enmtjnmSD0Y
GPZMVvtq95BTexxX34Xi21wWEbXT3A4Adc1Rdt4qL9HG130Xy2QR0Bxc2b5Kws
MtqcDeRrEeFn0InQM9nfbHKmX3LL1E8/znL7k6HNIDyCuUX9TtAyQ==
-----END RSA PRIVATE KEY-----
runsaï@DESKTOP-N6MPCBS MINGW64 ~/Downloads

```

```

-----END RSA PRIVATE KEY-----
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ec2-user@ec2-44-193-80-199.compute-1.amazonaws.com
The authenticity of host 'ec2-44-193-80-199.compute-1.amazonaws.com (44.193.80.199)' can't be established.
ED25519 key fingerprint is SHA256:DHGNhQHdTK/GrvrUKrPRrg7Ti+S26dr55j3jSzFp7mc.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:38: ec2-98-92-188-179.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-193-80-199.compute-1.amazonaws.com' (ED25519) to the list of known hos
ts.
#_

```


10. Configure VPC flow logs and store the logs in S3 and CloudWatch

VPC

Flow logs (2)				
Search				
☐	Name	Flow log ID	Traffic type	Destination type
☐	my-vpc-flowlogs-bucket-123	fl-0738d6ab90970fcb3	All	s3
☐	my-flogs	fl-0e33757a9cf6faada	All	cloud-watch-logs

S3

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

	Name	▲	Type	▼	Last modified	▼	Size
	AWSLogs/		Folder		-		

CloudWatch

CloudWatch Favorites and recents Ingestion Dashboards Alarms AI Operations GenAI Observability Application Signals (APM) Infrastructure Monitoring Loas	Log group "VPC-FlowLogs" has been created.									
	Log groups Data sources - new Summary									
	Get-set-go with unified data store									
	Log groups (1) By default, we only load up to 10,000 log groups. Filter log groups or try pattern search <table><tr><td>☐</td><td>Log group</td><td>Log class</td><td>Anomaly d...</td></tr><tr><td>☐</td><td>VPC-FlowLogs</td><td>Standard</td><td>Configure</td></tr></table>			☐	Log group	Log class	Anomaly d...	☐	VPC-FlowLogs	Standard
☐	Log group	Log class	Anomaly d...							
☐	VPC-FlowLogs	Standard	Configure							